

Technologies for E-Commerce

CAN302

**Department of Communications and Networking
Xi'an Jiaotong-Liverpool University (XJTLU)**

Lecture 5 – The history of e-commerce

Outline

1. Commerce, Business and Trade
2. Commerce and e-Commerce

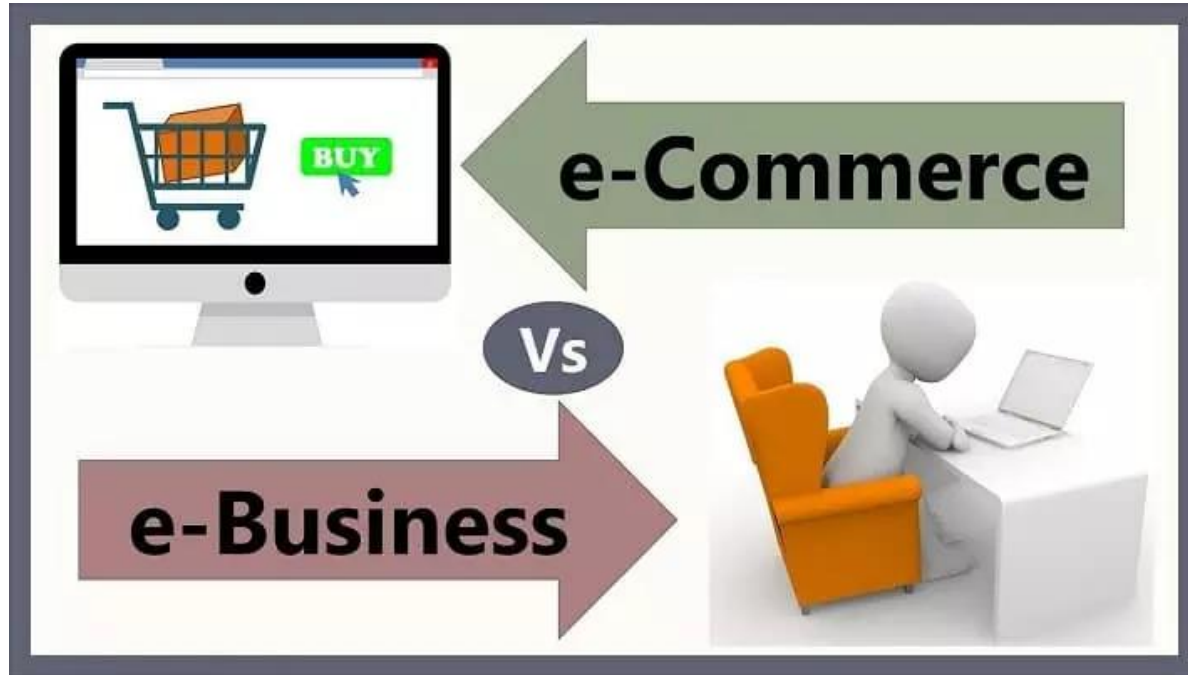


What's e-Commerce by your feelings?



- We have a very fast development in the digital technology in the past several decades.
- So the term changed very fast also.
- It is no strange you will see different definition in different materials and books.

Part1: Commerce, Business and Trade



Commerce vs. Business

- Commerce:

The **exchange** or buying and selling of commodities on a large scale **involving** transportation from place to place.

<https://www.merriam-webster.com/dictionary/commerce>

- Business:

The activity of **making**, buying, or selling goods or providing services in exchange for money

<https://www.merriam-webster.com/dictionary/business>

Commerce vs. Trade

- Trade:

The **business** of buying and selling or bartering commodities

<https://www.merriam-webster.com/dictionary/trade>

- Commerce:

The activity of buying and selling goods or providing services between organizations and/or individuals and **the activities of banks, insurers and other organizations**

Commerce vs. Trade

C o m m e r c e



Trade

**/ **

Home International



Services to trade

- Banking**
- Finance**
- Insurance**
- Transport**
- Communication**

The names of government department



中华人民共和国商务部

MINISTRY OF COMMERCE, PRC

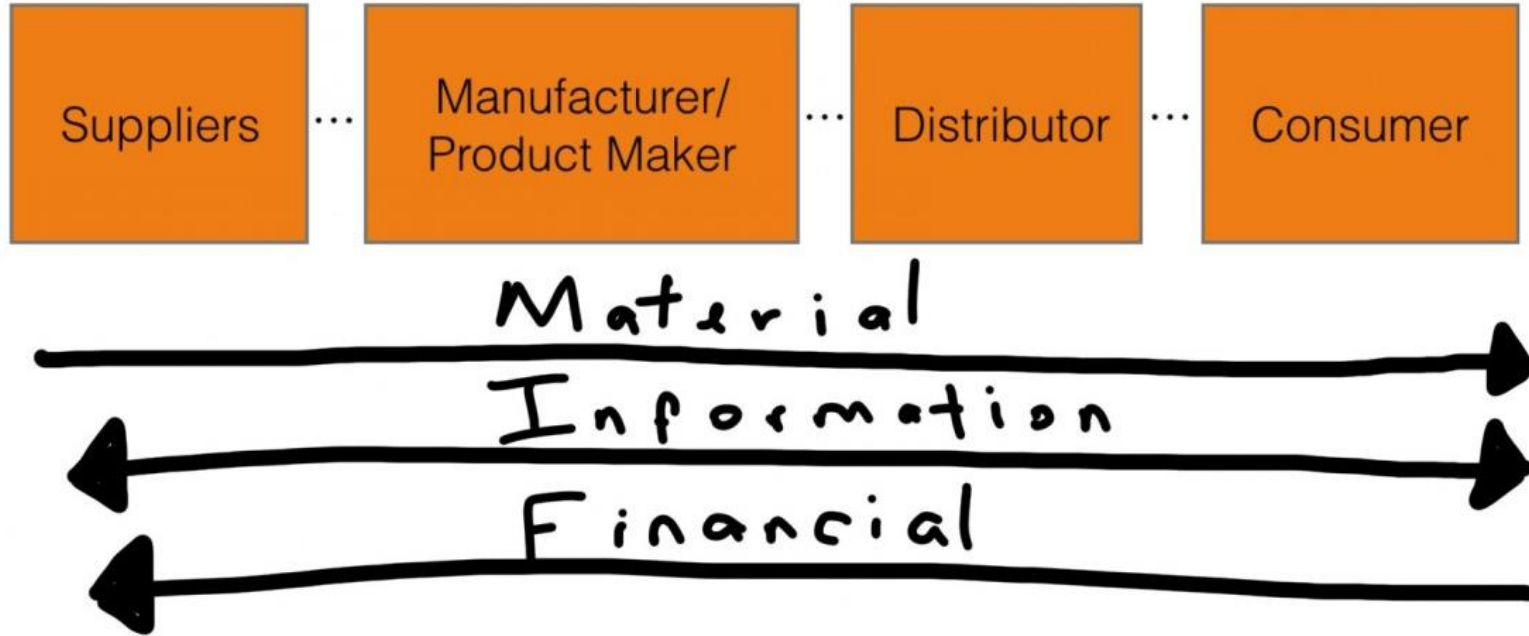


Department for
Business & Trade

Part2: Commerce and E-commerce



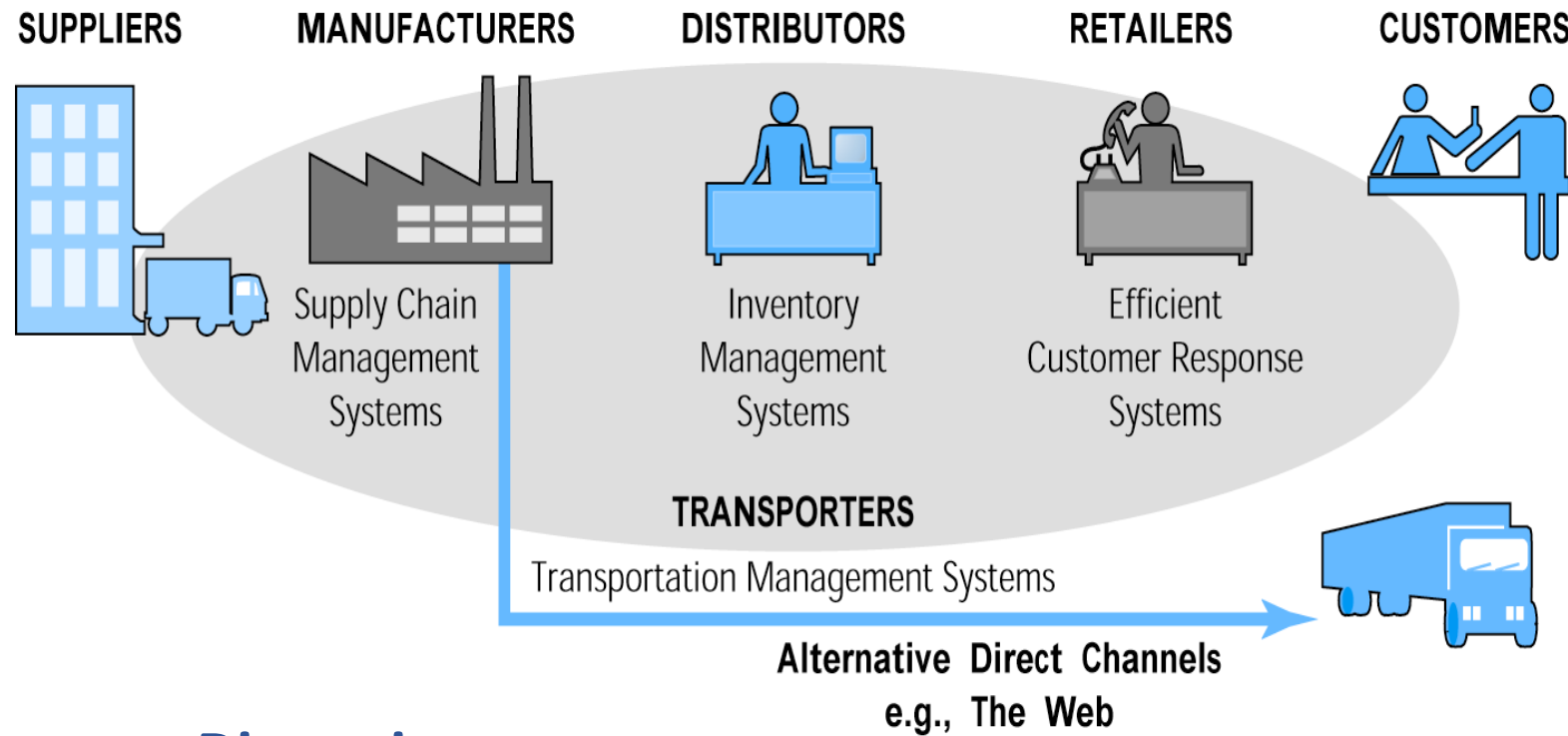
Three flows in commerce



<https://kpakpakpa.com/the-three-flows-of-supply-chain/>

A typical commerce session

Change the Goods **OWNERSHIP** - Information Cash/Money Product/Transport



- Search for goods
- Put the order
- Pay the order
- Receive the goods
- Customer services

Discussion:

- Which flow is digital now?

The ancient market format



- Nearby people gathering together or service provider wandering around

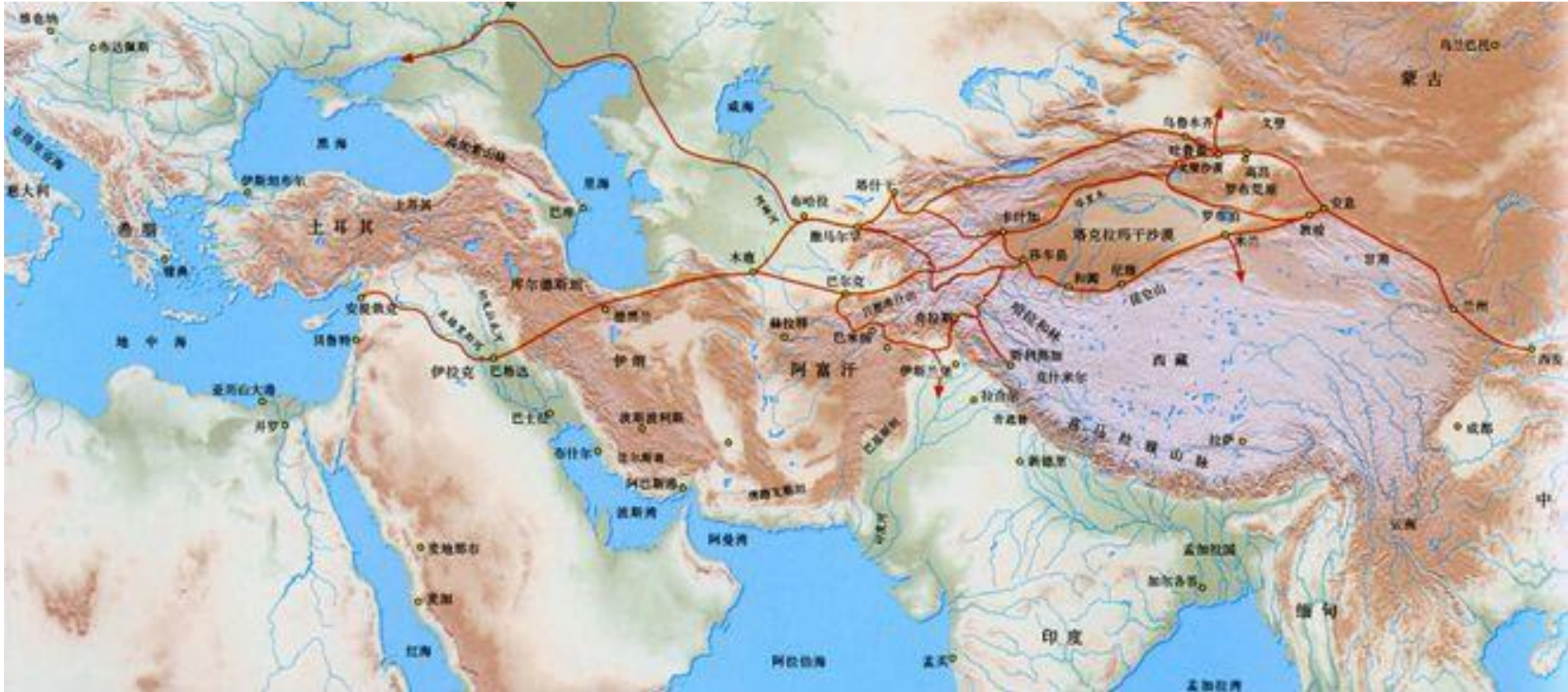
The ancient market format



东市买骏马
西市买鞍鞴
南市买辔头
北市买长鞭

- Fixed market for specific goods in different regions

The Silk Road



<https://www.163.com/dy/article/IS06JM3B055660VM.html>

The Age of Exploration



Department store



- A departmental store is a large establishment that sells a wide range of products organized into distinct departments in order to satisfy nearly every customers' requirements under one roof.
- It is divided into several departments, each of which focuses on a specific type of product, including clothing, electronics, furniture, and more

<https://www.geeksforgeeks.org/departmental-stores-features-types-advantages-disadvantages/>

Super Market



- Part of daily life

Advertisement



- Paper and TV based advertisement
- Push the **information** to potential customers

TV shopping

LIVE

DALLAS, TEXAS : 214 - 304 - 1277
HOUSTON, TEXAS : 714 - 894 - 5299

Live on 57.11 (Los) | Galaxy 19 | Youtube | VStarTV.net | VStar TV App (iPhone & Android)

VIET USA
HOME SHOPPING

714-352-9483
9393 Bolsa Ave - Suite D, Westminster, CA
(GÓC BOLSA & BUSHARD)
Chi Nhánh 1

888 Home Shopping
626-420-3946
9567 Garvey Ave #17-18, South El Monte, CA
(TRONG KHU CROWN PLAZA)
Chi Nhánh 2

- A kind of advanced advisement
- Put the order over a telephone directly

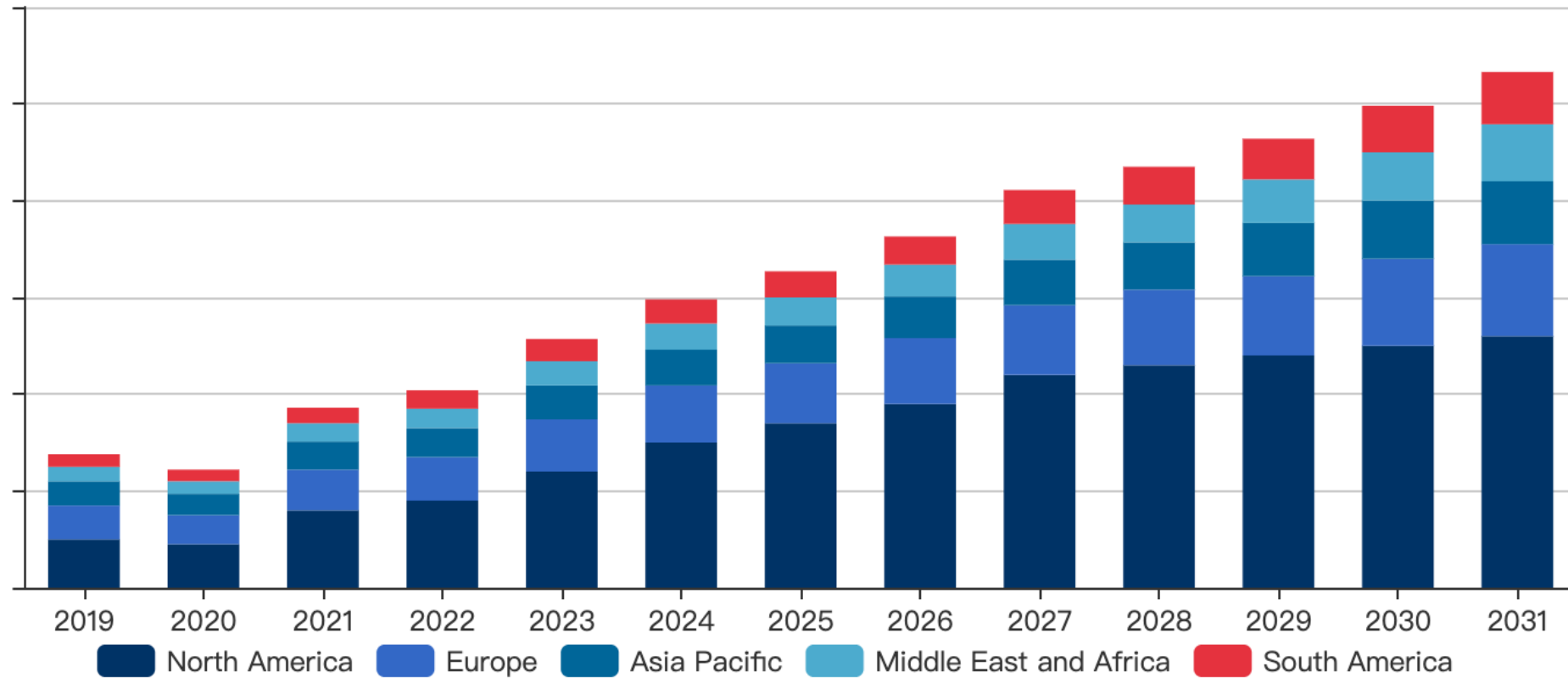
A definition of e-Commerce

- **Electronic commerce (EC)**

Electronic enabled commercial transactions between and among organizations and individuals

- Does TV shopping a kind of e-commerce?

TV shopping market



- It is more popular in US
- It is still active!

<https://www.cognitivemarketresearch.com/tv-shopping-market-report>

Extended definition of e-Commerce

- **Electronic commerce (EC)**

Electronic enabled commercial transactions between and among organizations and individuals

- **Internet commerce (Web commerce)**

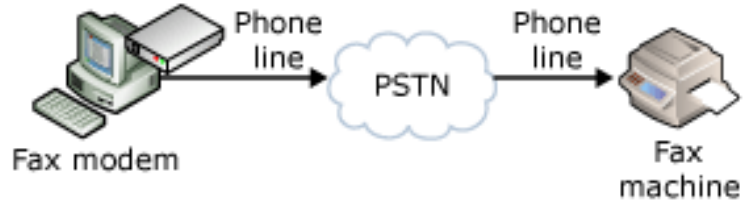
The process of buying, selling, or exchanging products, services, or information via **Internet**

FAX or facsimile

Traditional fax machine



Fax modem



- Fax is also an electronic enabled communication method.
- While you may see it is not mentioned in e-Commerce. **Why?**

OECD definition of e-commerce

Table 4.1. The 2025 OECD definition of e-commerce

Definition of e-commerce	Guidelines for interpretation
An e-commerce transaction is the sale or purchase of goods or services, conducted over computer networks by methods specifically designed for the purpose of receiving or placing of orders. The goods or services are ordered by those methods, but the payment and the ultimate delivery of the goods or services do not have to be conducted online. An e-commerce transaction can be between enterprises, households, individuals, governments, and other public or private organisations.	Include: Orders made via webpages, apps, extranets, and Electronic Data Interchange (EDI). The type is defined by the method of making the order. Exclude: Orders made by telephone calls, facsimiles, on-premise mechanisms (including but not limited to kiosks or QR codes), manually typed messages (including but not limited to e-mails).

- The Organisation for Economic Co-operation and Development (OECD) is an international organisation that works to build better policies for better lives. Electronic enabled commercial transactions between and among organizations and individuals.

https://www.oecd.org/content/dam/oecd/en/publications/reports/2025/10/the-2025-oecd-definition-of-e-commerce-and-guidelines-for-interpretation_3e7e5a61/2254f1de-en.pdf

The 1st product sold online



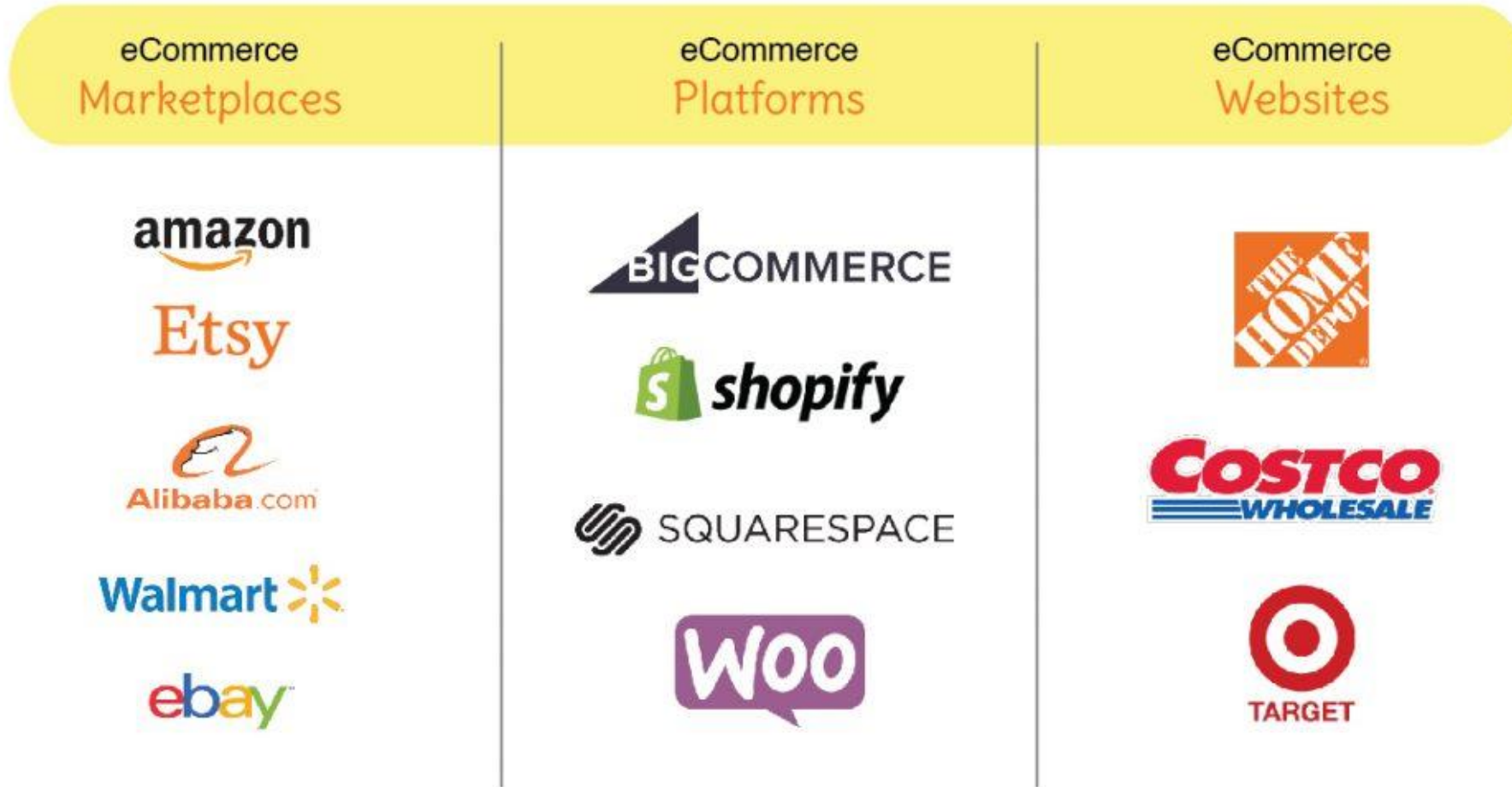
- 11th Aug, 1994, a CD was sold through www.netmarket.com
- \$12.48+shipping
- The credit card No. was encrypted in the purchasing process.

<https://www.smithsonianmag.com/smart-news/what-was-first-thing-sold-internet-180957414/>

Mainstream e-commerce platforms



Marketplace and Platforms



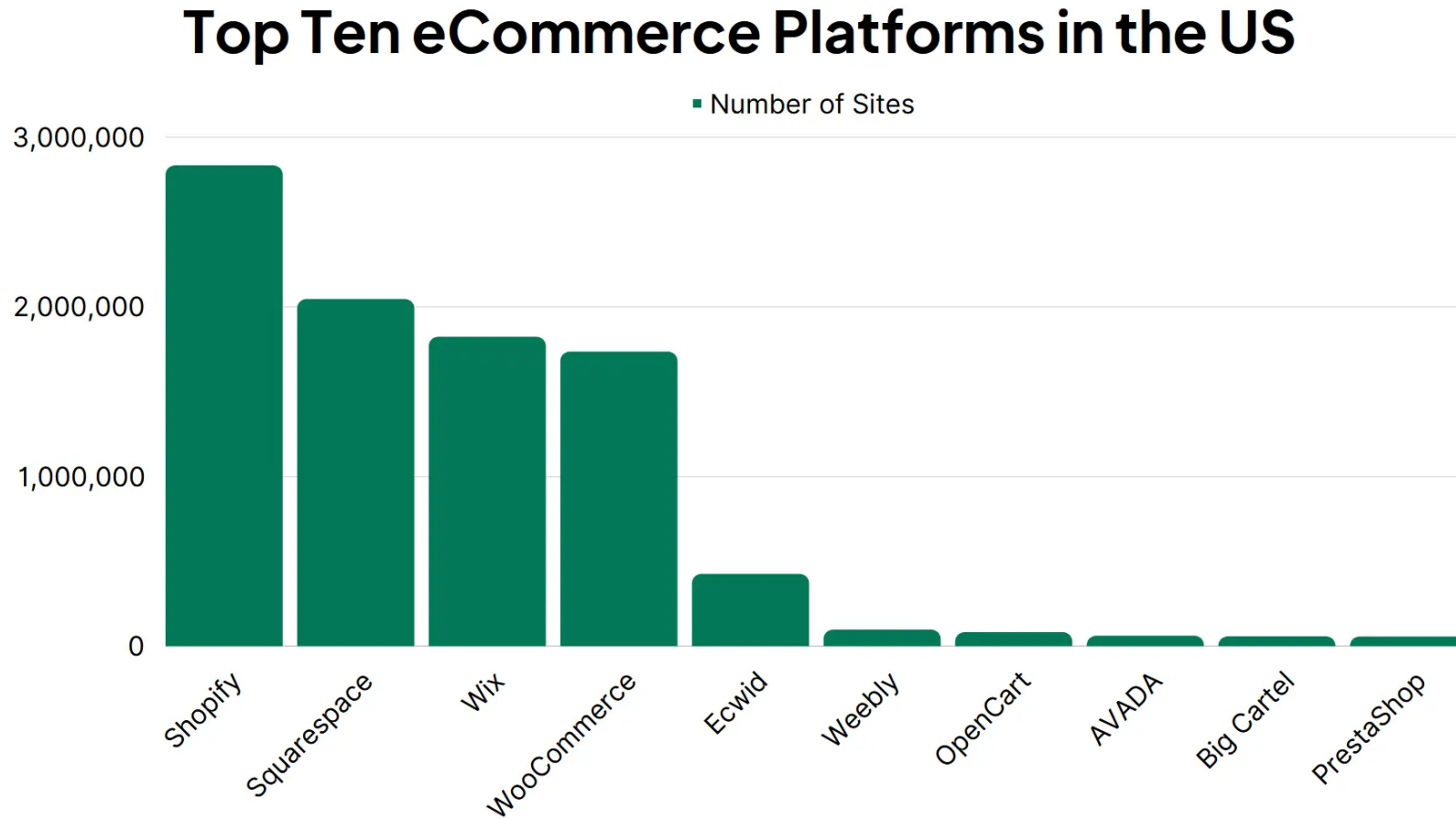
<https://translatebyhumans.com/blog/history-and-evolution-of-ecommerce/>

A glance at shopify



<https://millionify.com/features-and-benefits-of-shopify/>

So many sites ...

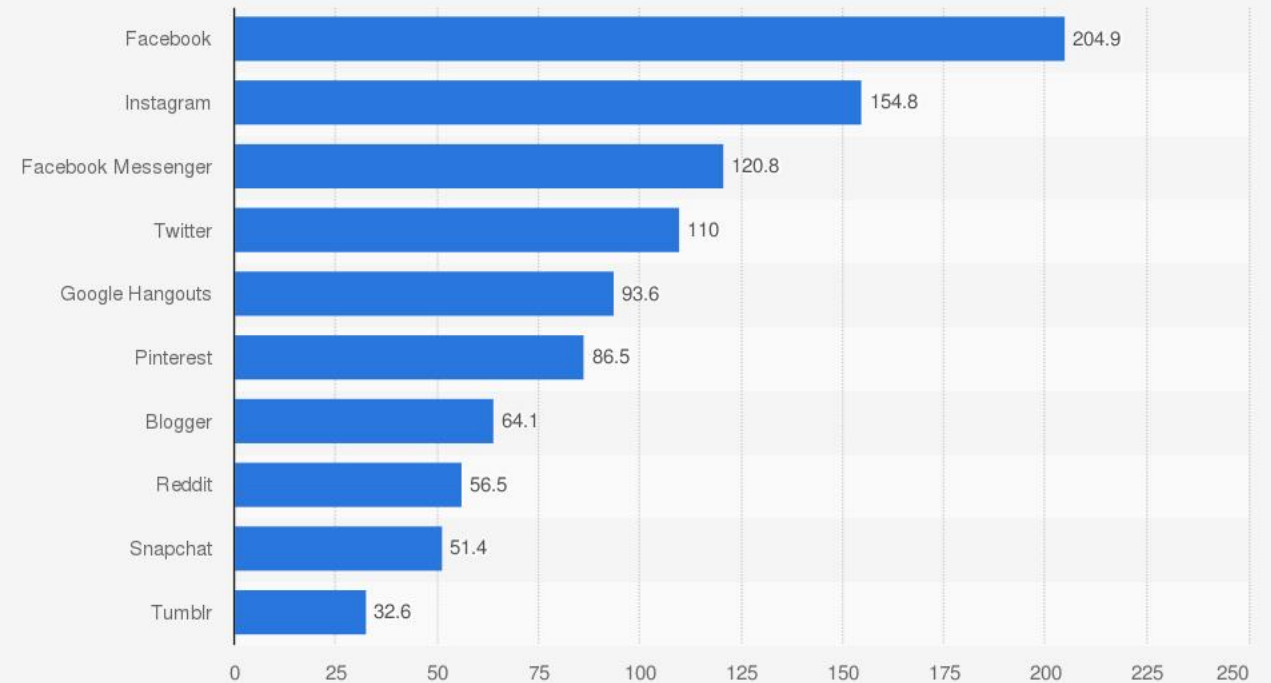


<https://www.mobiloud.com/blog/ecommerce-platform-market-share-usa>

Website Traffic



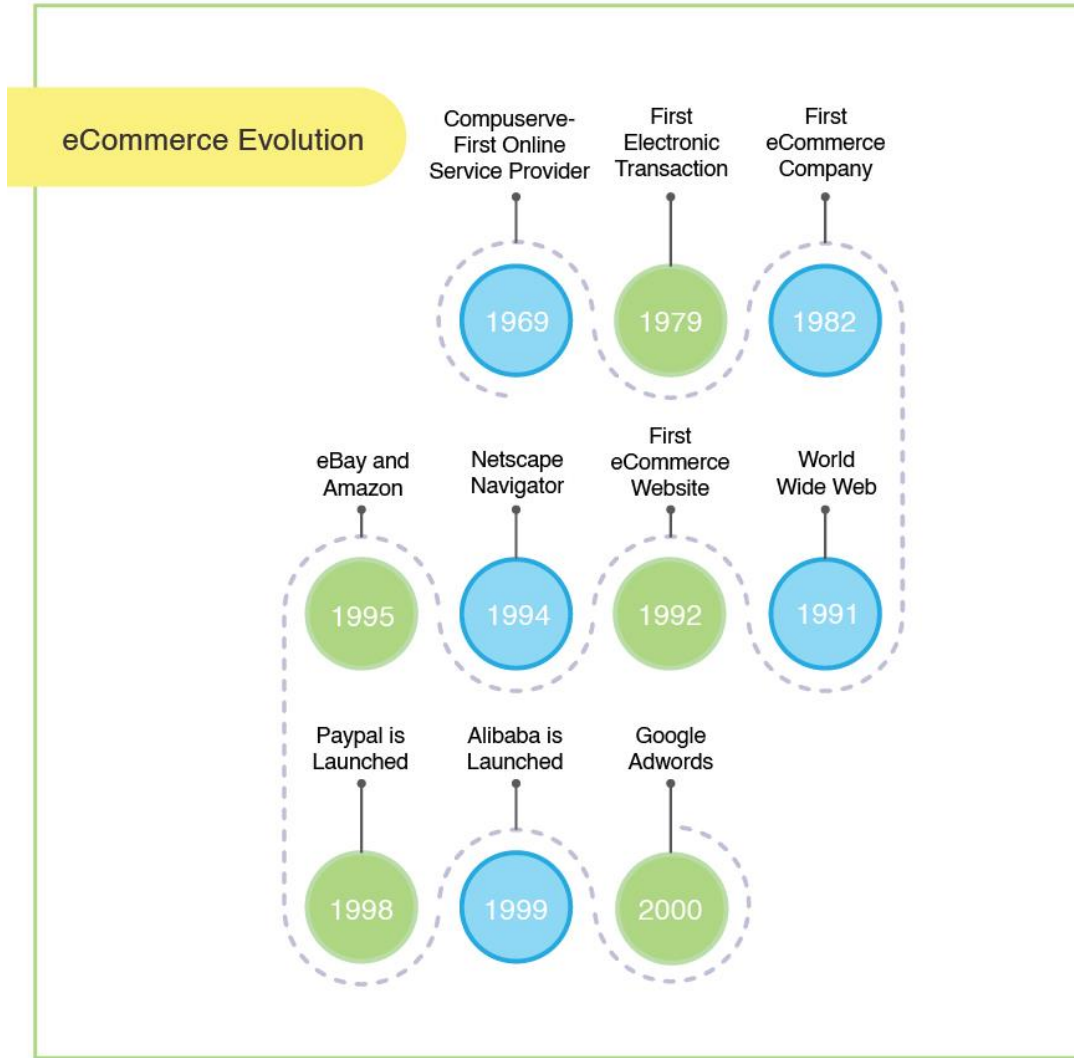
BEST E-COMMERCE TRAFFIC SOURCES



- Traffic always means money!

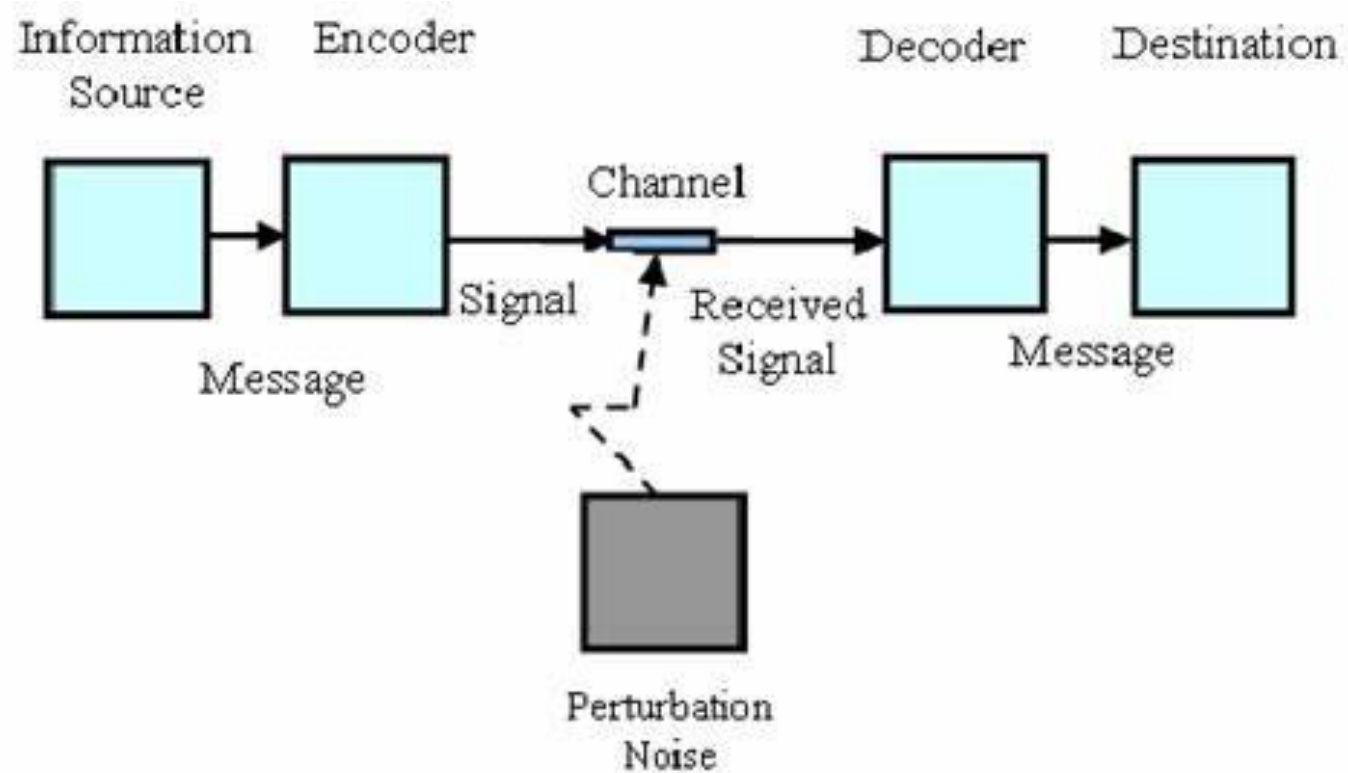
<https://www.targetedwebtraffic.com/what-are-the-best-website-traffic-sources/>

E-Commerce Evolution



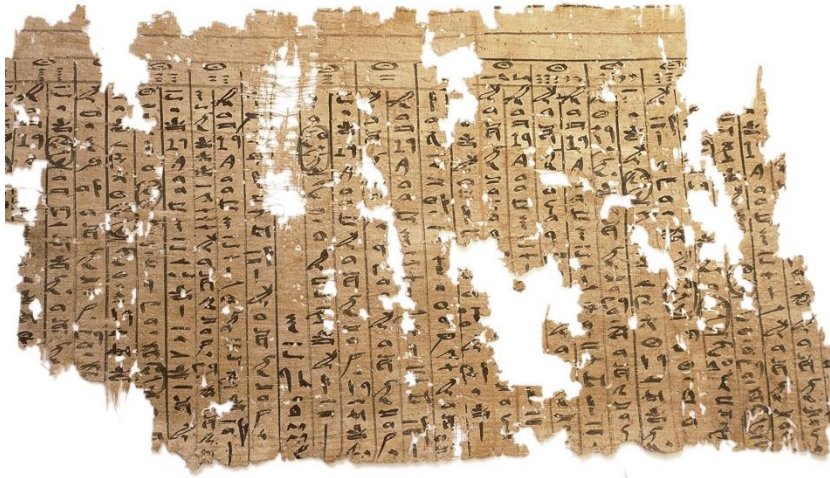
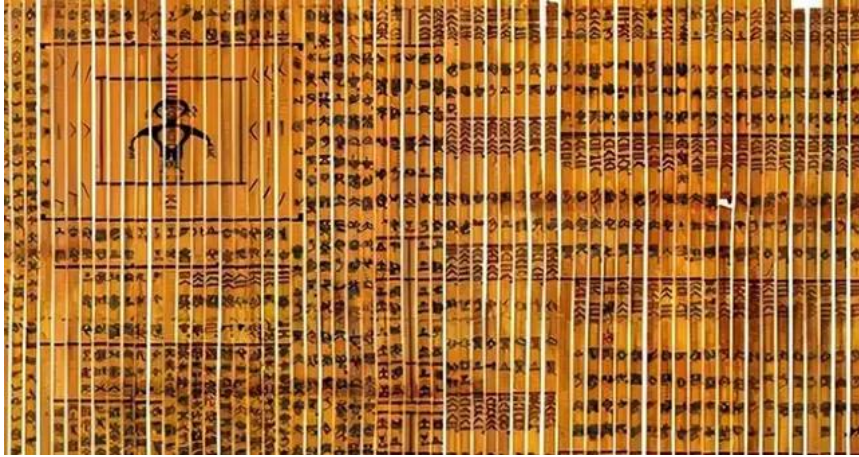
- 1969: CompuServe (a company) – first online service provider – is launched.
- 1979: Michael Aldrich invents electronic shopping and makes the first **electronic** transaction.
- 1982: World's first eCommerce company, **Boston Computer Exchange**, is started.
- 1991: The **world wide web** is invented.
- 1992: First eCommerce website – an online bookstore – Book Stacks Unlimited is launched.
- 1994: Netscape Navigator – an online web browser – is launched.
- 1995: eBay and Amazon are launched.
- 1998: **Paypal** is launched, and it revolutionizes online payment processes.

Underneath the e-Commerce Evolution



- The evolution of communication technologies

Ancient communication system

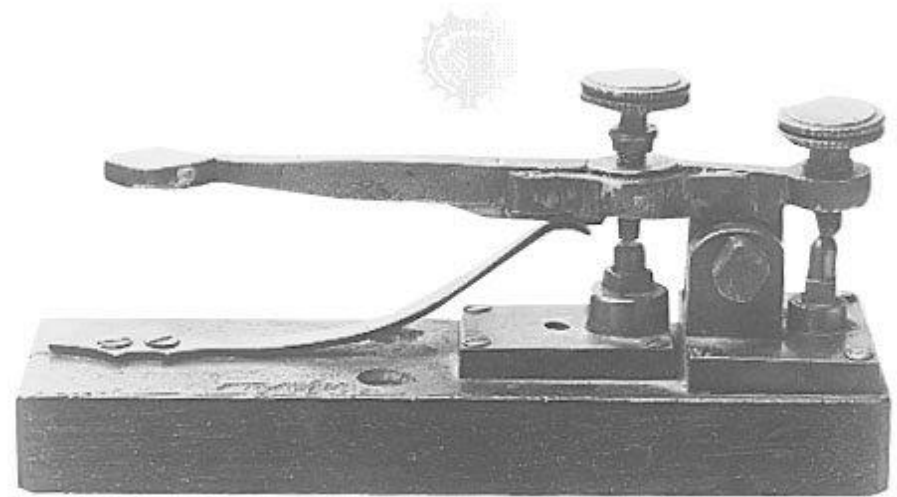


Ancient communication system



- Optical Telegraph in France

Invention of electronic telegraphy



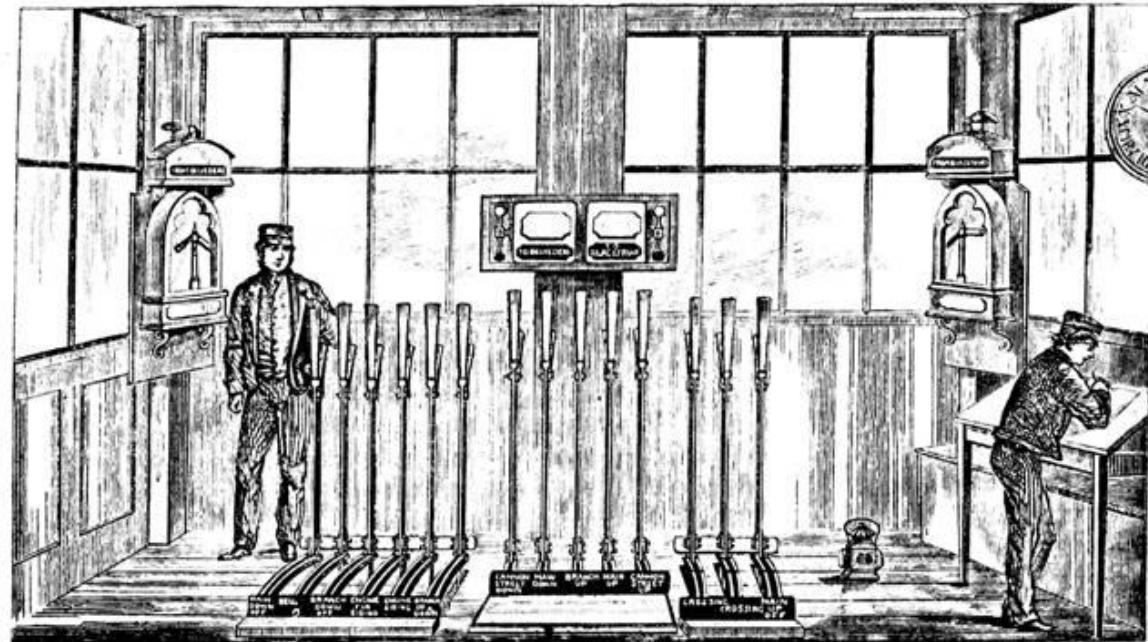
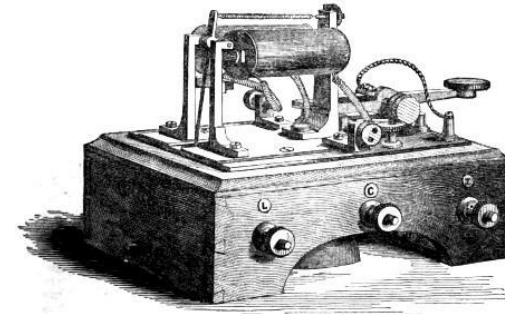
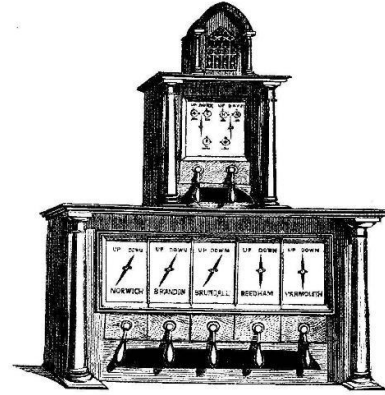
<https://baike.baidu.com/item/电报/110262>

<https://historyhome.co.uk/peel/railways/puborder.htm>

<https://www.britannica.com/technology/telegraph>

<https://wenku.baidu.com/view/1b21be23dbef5ef7ba0d4a7302768e9950e76ed2.html>

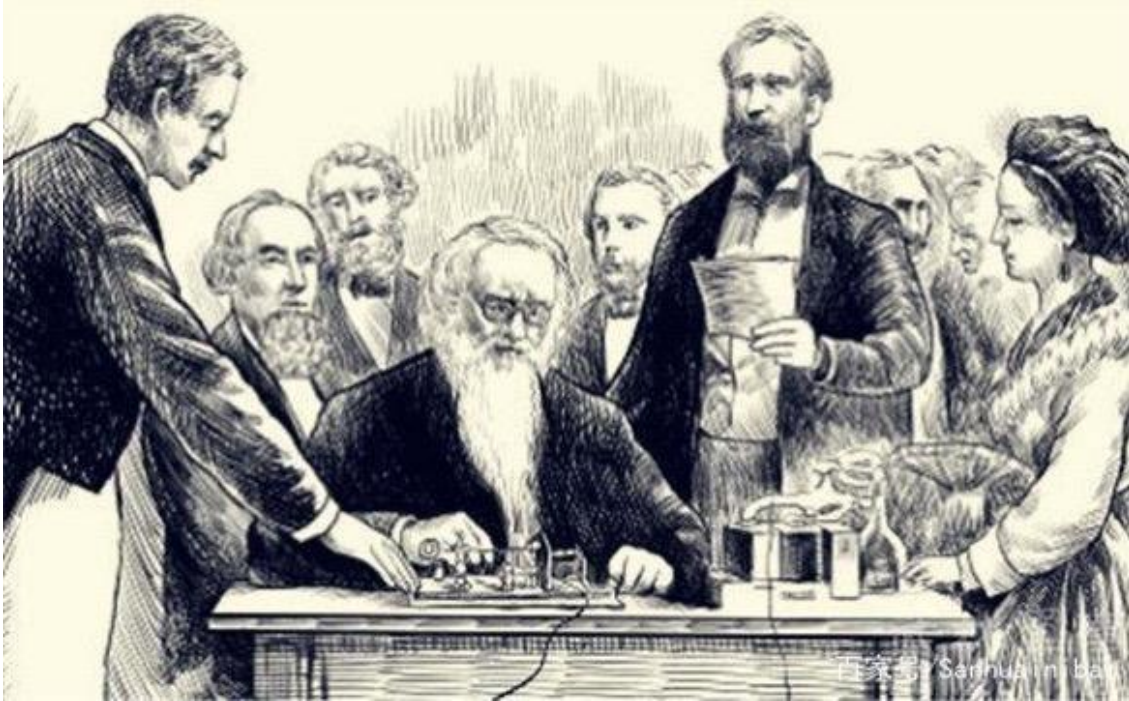
Big BEN and telegraphy



- Telegraph, from the Greek “tele”, distant, and “graphos”, writing

<https://distantwriting.co.uk/railwaysignaltelegaphy.html>

Morse code



- Physical layer

A	• —	M	— —	Y	— • — —
B	— • • •	N	— •	Z	— — • •
C	— • — •	O	— — —	1	• — — — —
D	— • •	P	• — — •	2	• • — — —
E	•	Q	— — — •	3	• • • — —
F	• • — •	R	• — •	4	• • • • —
G	— — — •	S	• • •	5	• • • • •
H	• • • •	T	—	6	— • • • •
I	• •	U	• • —	7	— — — • •
J	• — — — —	V	• • • —	8	— — — — • •
K	— • —	W	• — —	9	— — — — •
L	• — • •	X	— • • —	0	— — — — —

- Coding

Telegraphy money order

Certificate of the Issue of a
British Money Order

SS 42 986

Payable at Worcester
To Mrs R Russell

	£	s.	d.
Amount of Order.....	35	-	-
Money Order Charges.....		1	2
Telegraph Charges.....		1	4
Total	35	2	6

This form is of no monetary value but should be retained by the sender for purposes of reference.

Issuing Officer.

Wt. 9547-McC-(6)

IBCC Digital Archive

□□□□□□ 中国人民邮政汇款通知

请按照背面注意事项携带本通知和本人有效证件名章到局取款

汇9965 扶绥532100

收汇局

收汇员 谢天明

收汇日期 89.03.28.03

兑付员 廖永新

接收专号 03038

邮政编码

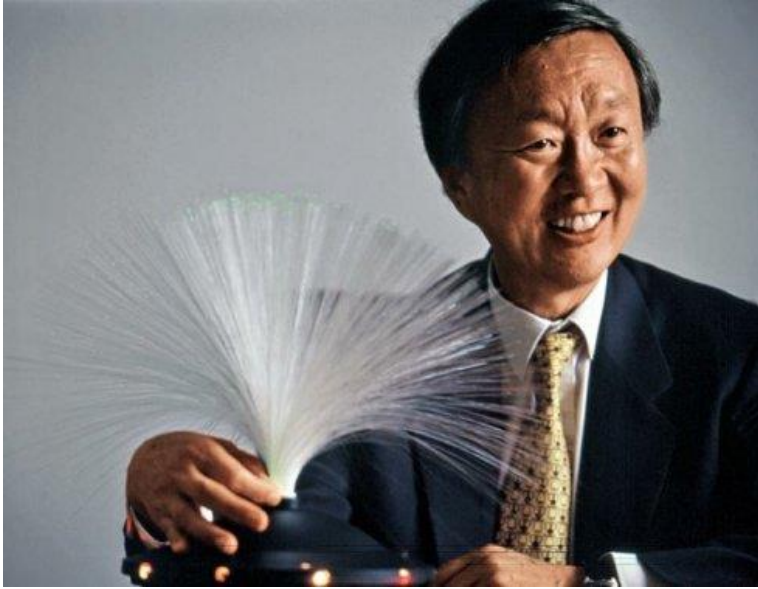
汇款金额	人民币 (大写) 叁佰圆正
收款人详细地址	天等县人民医院
收款人姓名	黄贵宁
汇款人详细地址	扶绥县中东镇粮所
汇款人姓名	凌裕安
要否航空	

- Telegraphy is not only for message but for money order.
- Post office operated the money transfer for a very long time.

Telephone by Bell



Optical fiber communication started here



Dielectric-fibre surface waveguides for optical frequencies

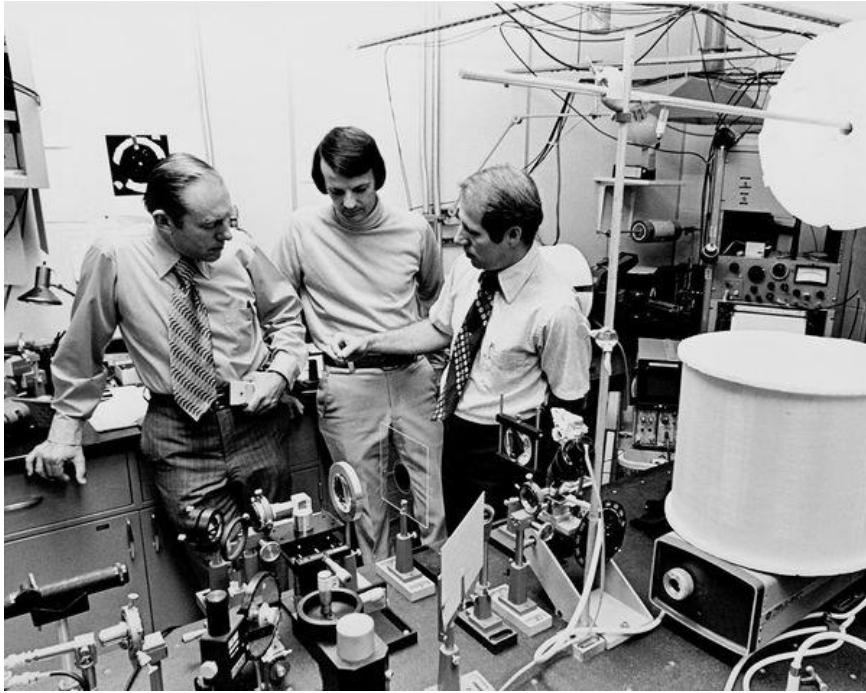
K.C. Kao and G.A. Hockham

Indexing terms: Optical fibres, Waveguides

Abstract: A dielectric fibre with a refractive index higher than its surrounding region is a form of dielectric waveguide which represents a possible medium for the guided transmission of energy at optical frequencies. The particular type of dielectric-fibre waveguide discussed is one with a circular cross-section. The choice of the mode of propagation for a fibre waveguide used for communication purposes is governed by consideration of loss characteristics and information capacity. Dielectric loss, bending loss and radiation loss are discussed, and mode stability, dispersion and power handling are examined with respect to information capacity. Physical-realisation aspects are also discussed. Experimental investigations at both optical and microwave wavelengths are included.

- The theoretical prediction made in 1966
- Dr Kao was not the first one for optical fiber but but the first one made it work for communication!

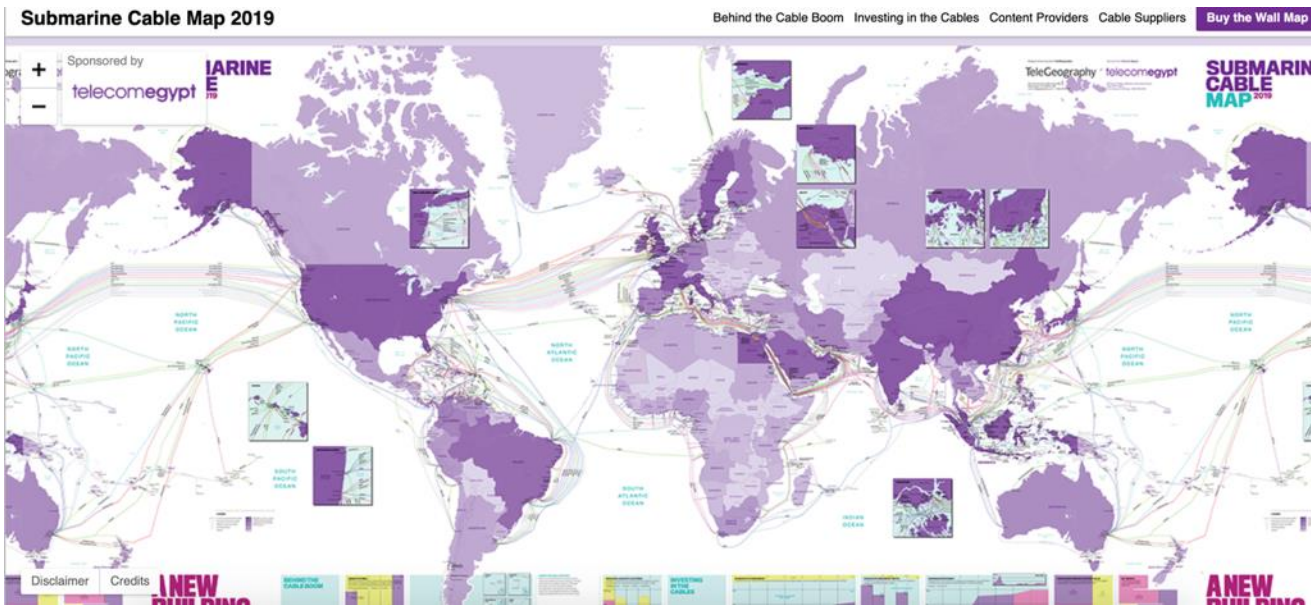
Purification is the secret



- In 1970, **Corning** scientists Dr. Robert Maurer, Dr. Peter Schultz, and Dr. Donald Keck developed a **highly pure** optical glass that transmitted light signals over long distances.

<https://www.corning.com/worldwide/en/markets/Optical-Communications-Market/streamlined-connectivity/timeline-1970.html>

Link the whole world!



Telecom

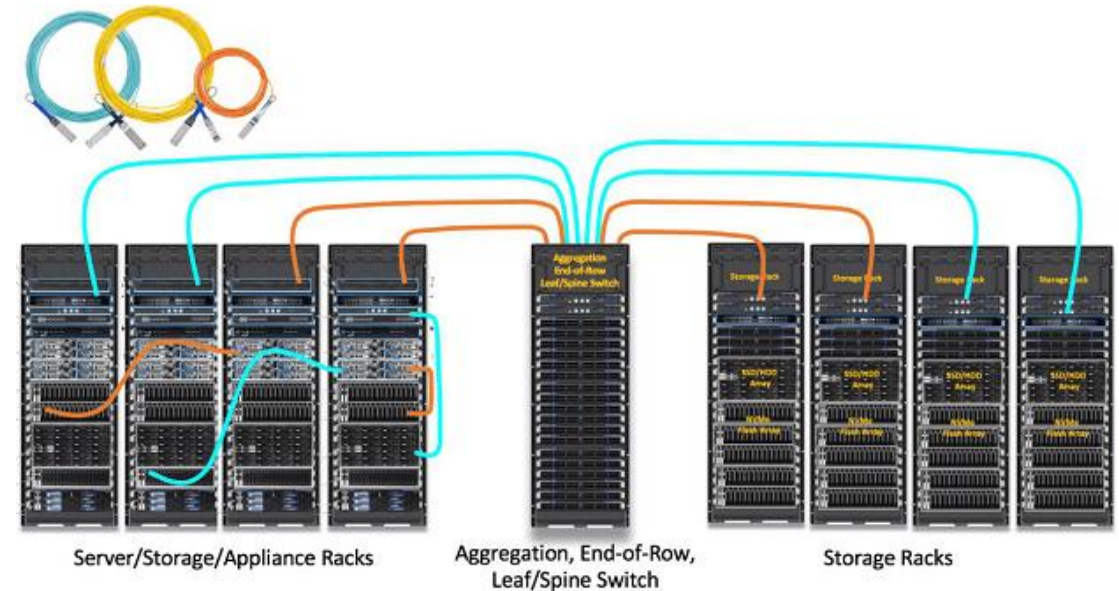
Sea systems:

> 400 systems

> 1,200,000 km

Land systems:

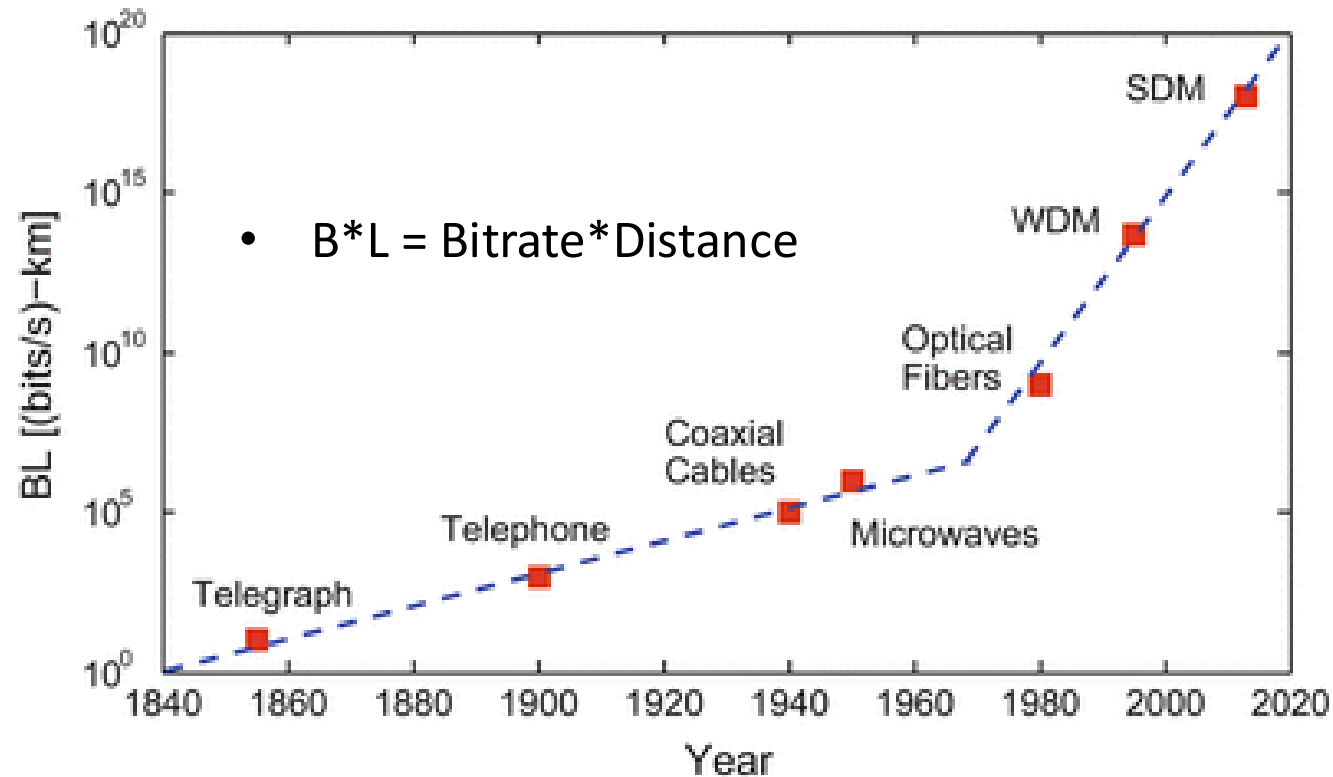
China > 47,500,000 km



Data center

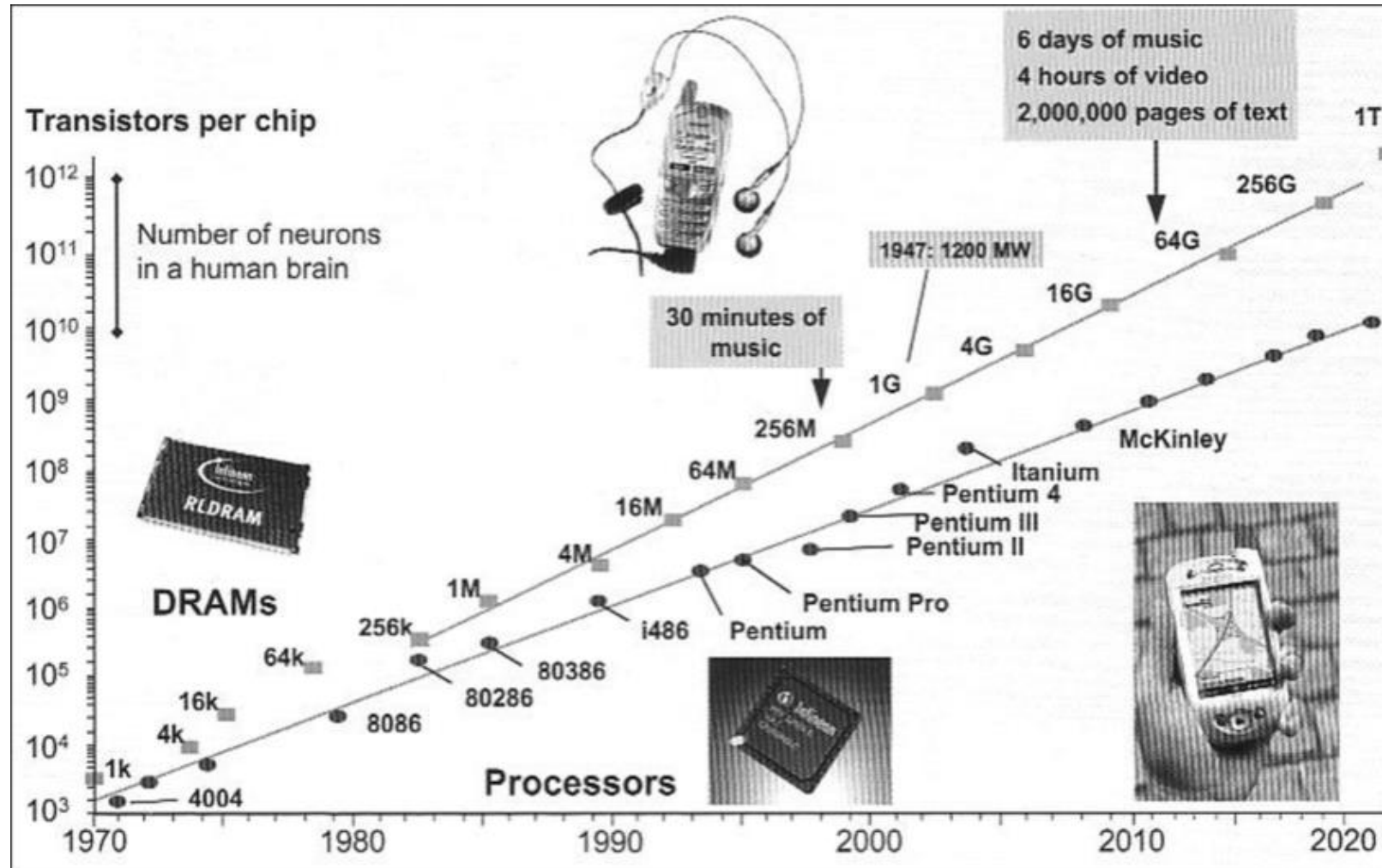
The new driving force

The B*L improvements



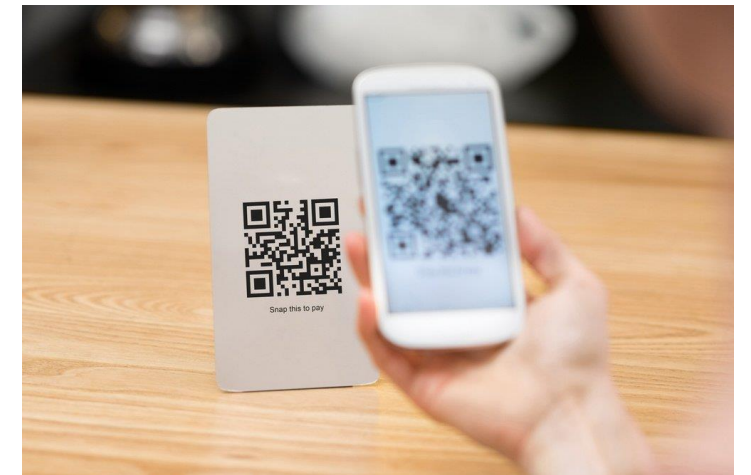
- The increase of $B*L$ always results in application revolutions
- We are chasing both of the high bandwidth and long distance
- Sometime, we need to make a compromise of them

Underneath the e-Commerce Evolution



- The evolution of terminal devices

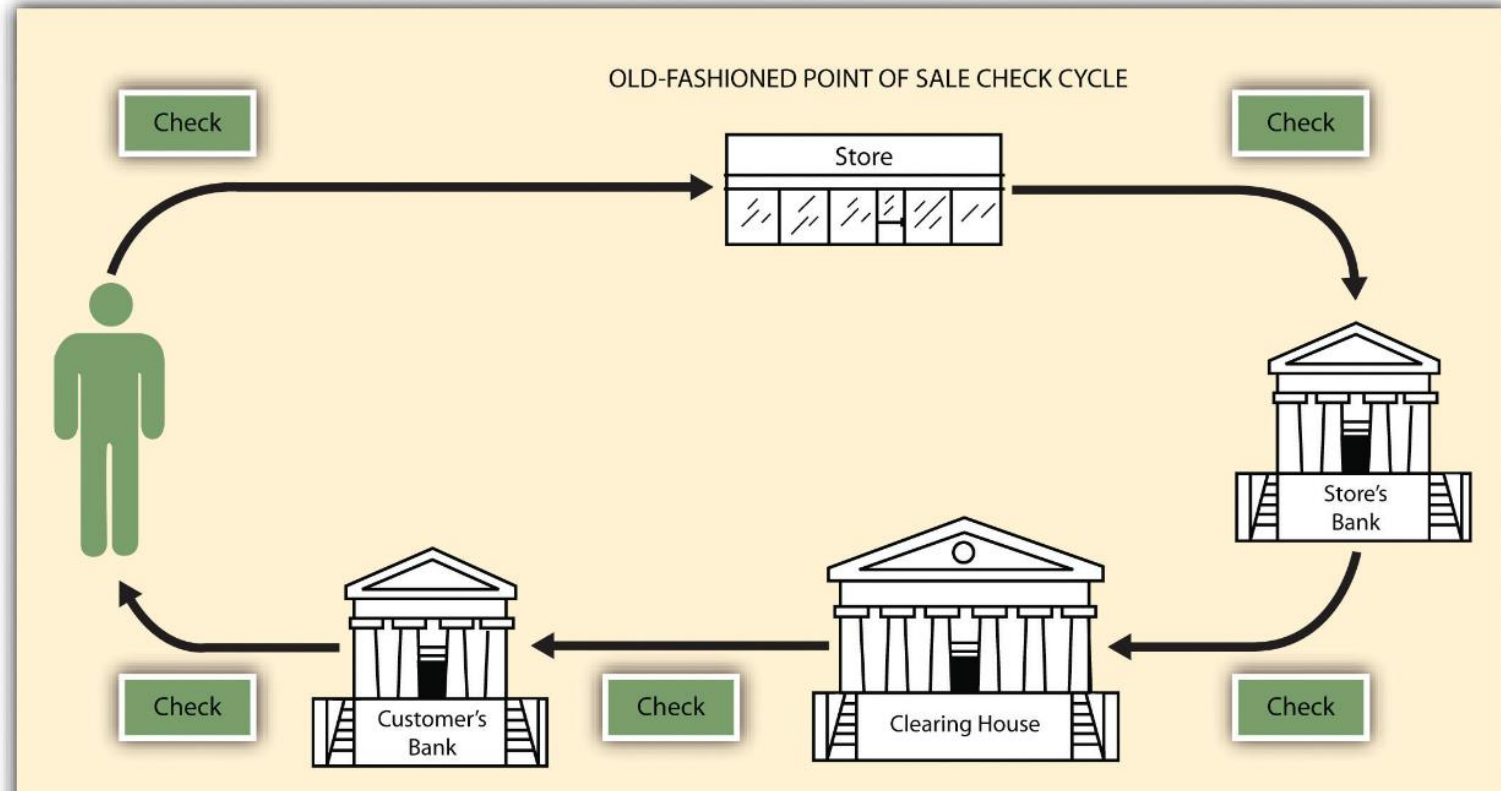
The revolution of payment



Discussion: which one is the first e-payment?

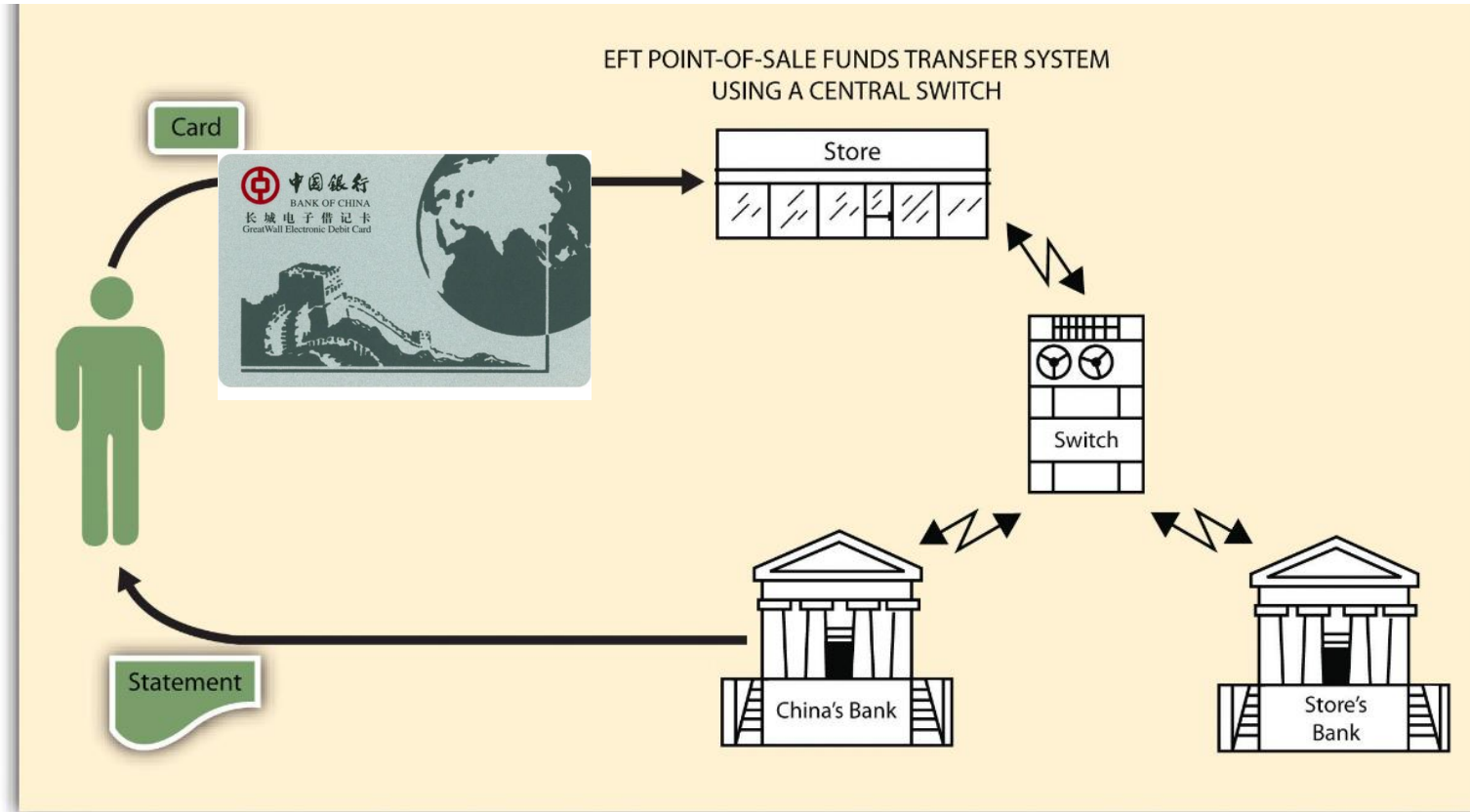
<https://cj.sina.com.cn/article/detail/6137509981/195123>

Why it is Check?



《Catch me if you can》

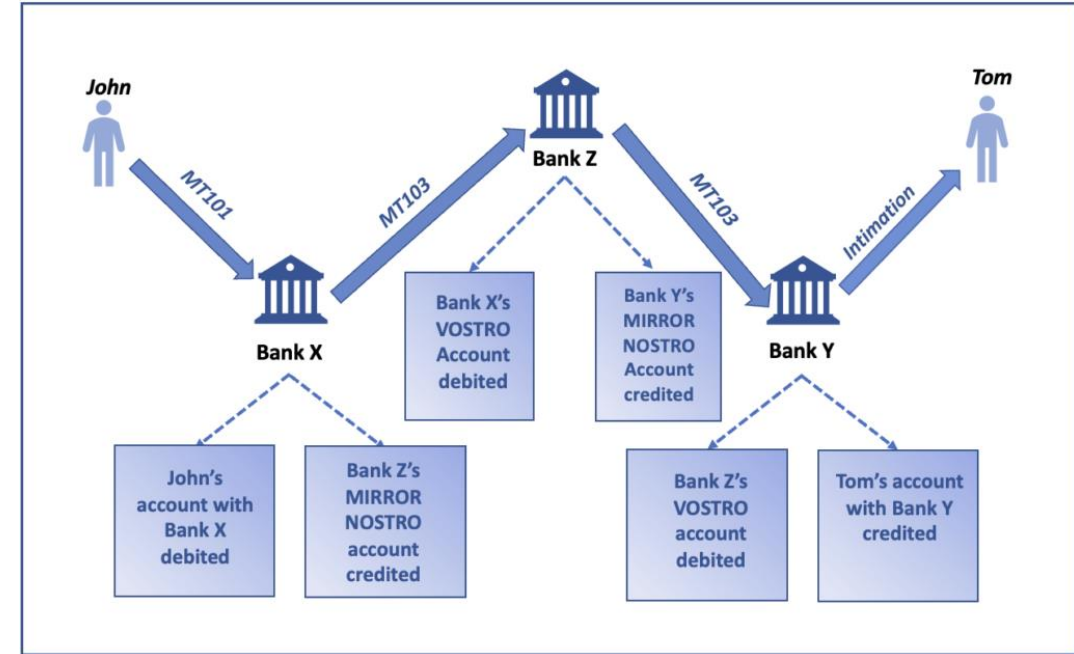
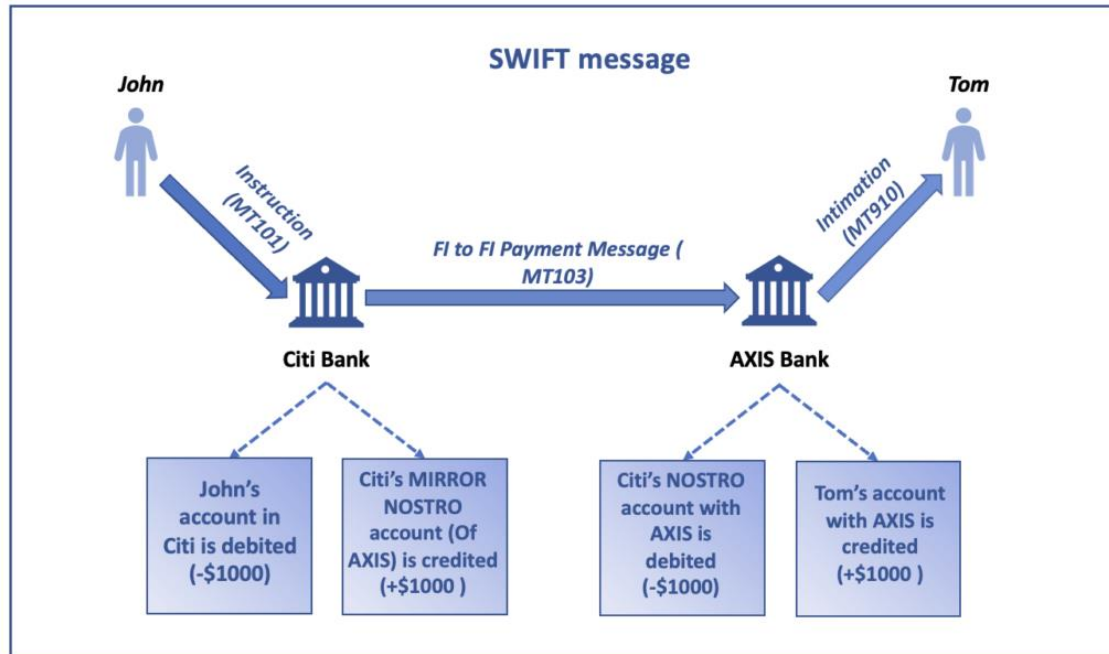
EFT: electronic fund transfer



https://saylordotorg.github.io/text_advanced-business-law-and-the-legal-environment/s20-02-electronic-funds-transfers.html

《Entrapment》

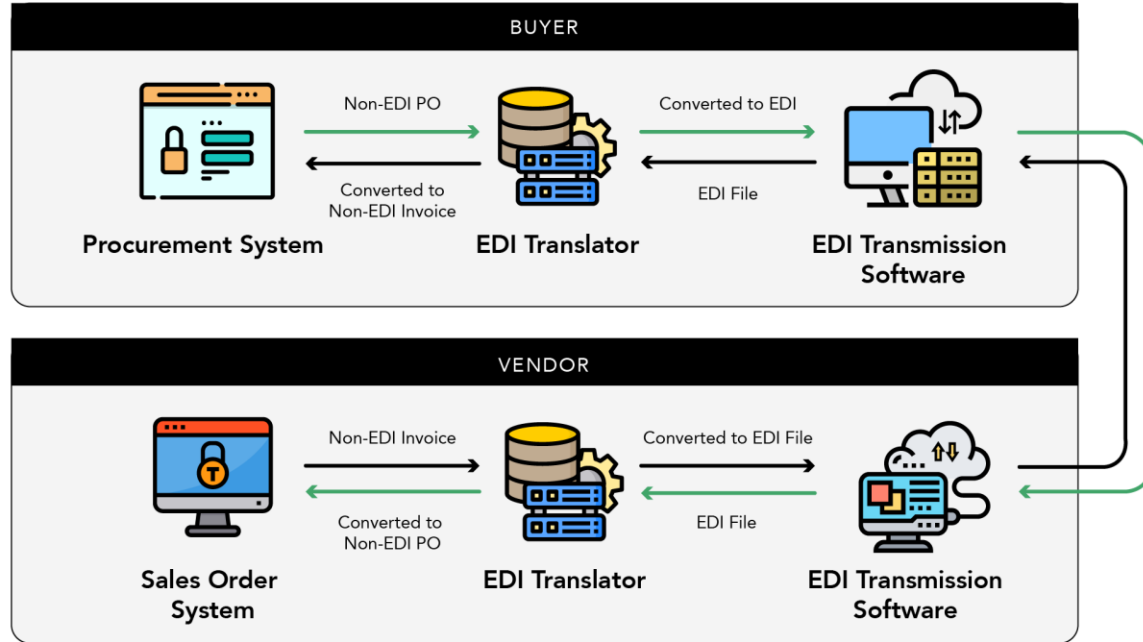
SWIFT: an important global foundation



- The Society for Worldwide Interbank Financial Telecommunications (**SWIFT**) system powers most international money and security transfers
- It is also very powerful like an **financial nuclear bomb**

<https://www.paymentsdomain.com/2021/11/21/swift-mt-serial-and-cover-method/>

EDI: Electronic Data Interchange



- EDI extend the e-commerce **beyond** e-payment.
- EDI is the intercompany communication of **business documents** in a standard format.
- The simple definition of EDI is a standard electronic format that replaces paper-based documents such as **purchase orders** or **invoices**.

<https://www.digitalcommerce360.com/2020/06/01/why-edi-still-has-a-big-role-to-play-in-b2b-e-commerce/>
<https://wenku.baidu.com/view/d585c486b9f67c1cfad6195f312b3169a451ea8c.html>
<https://www.comarch.com/trade-and-services/data-management/news/why-edi-is-still-in-use/>
<https://wenku.baidu.com/view/19e13edbf6ec4afe04a1b0717fd5360cba1a8dec.html>
<https://www.ibm.com/topics/edi-electronic-data-interchange>

A EDI sample

Paper Purchase Order

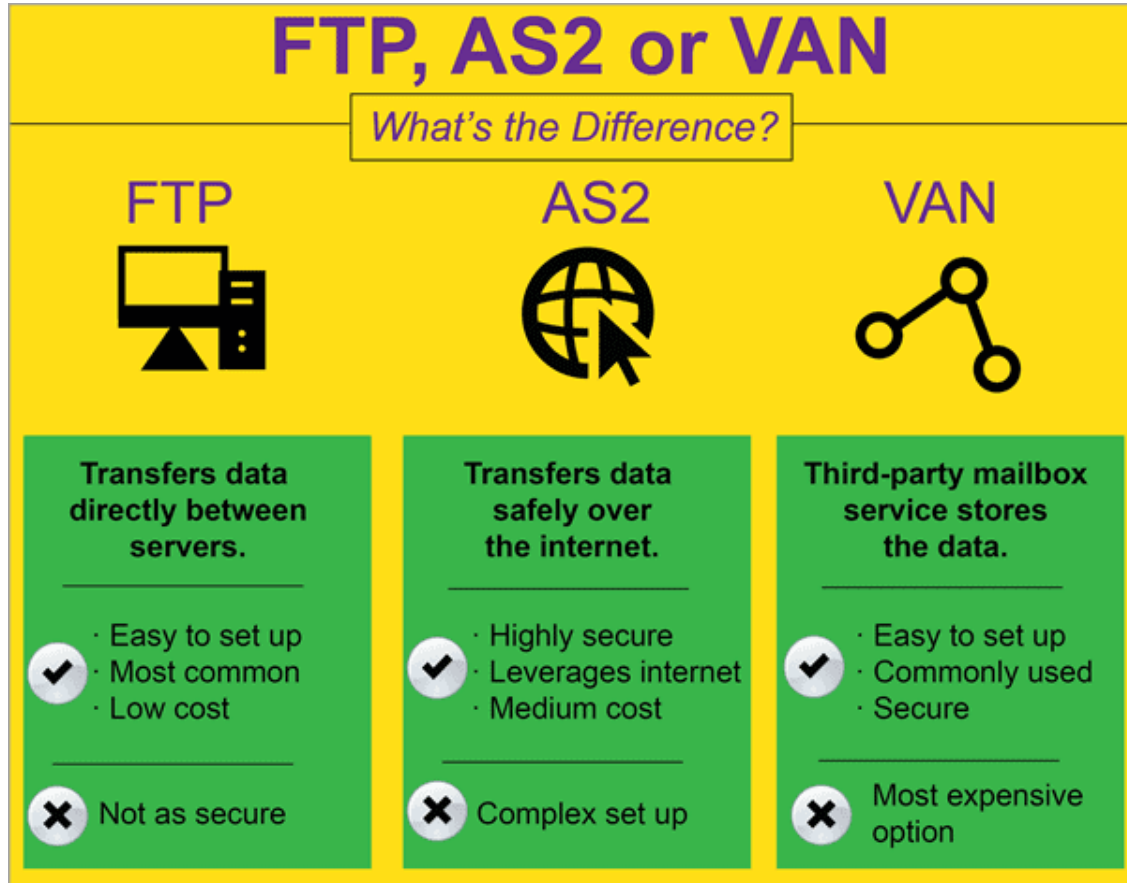
Purchase Order				
Sold To: XYZ Company		PO Number: 4768		
123 Main Street				
Fairview, CA 94168		PO Date: 9/30/2012		
Item No.	Quantity	Unit of Measure	Price	Product ID
1	100	EA	27.65	331896-42
Total Items: 1		Total Quantity: 100		

ANSI EDI Purchase Order

```
ST*850*540001■  
BEG*00*SA*4768*65*20120930■  
N1*SO*XYZ Company■  
N3*123 Main Street■  
N4*Fairview*CA*94168■  
PO1*1*100*EA*27.65**VN*331896-42■  
CTT*1*100■  
SE*8*54001■
```

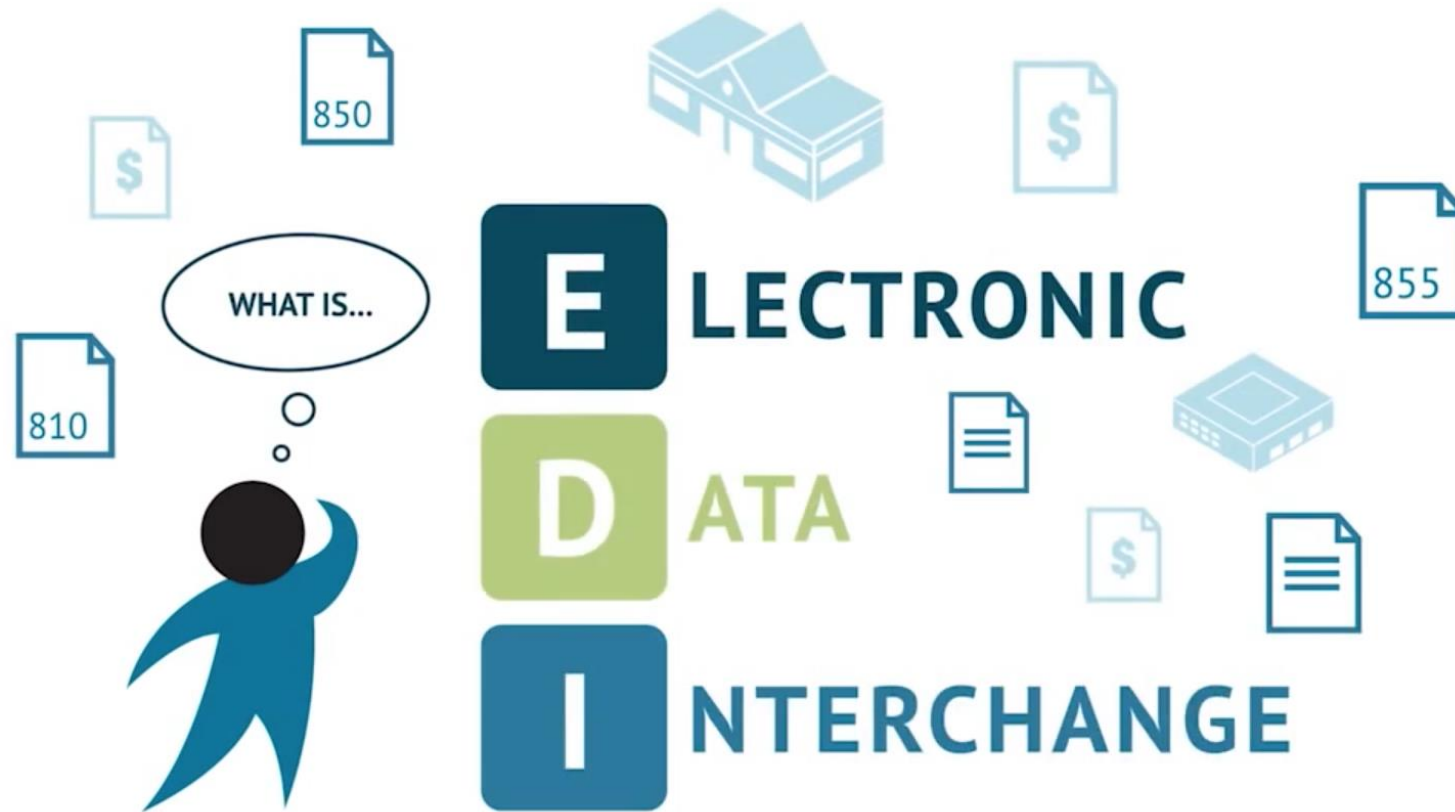
- Similar to ETF, exchange the key data with a fixed standard format.
- But extend to **purchase orders** and **invoices**.

EDI types

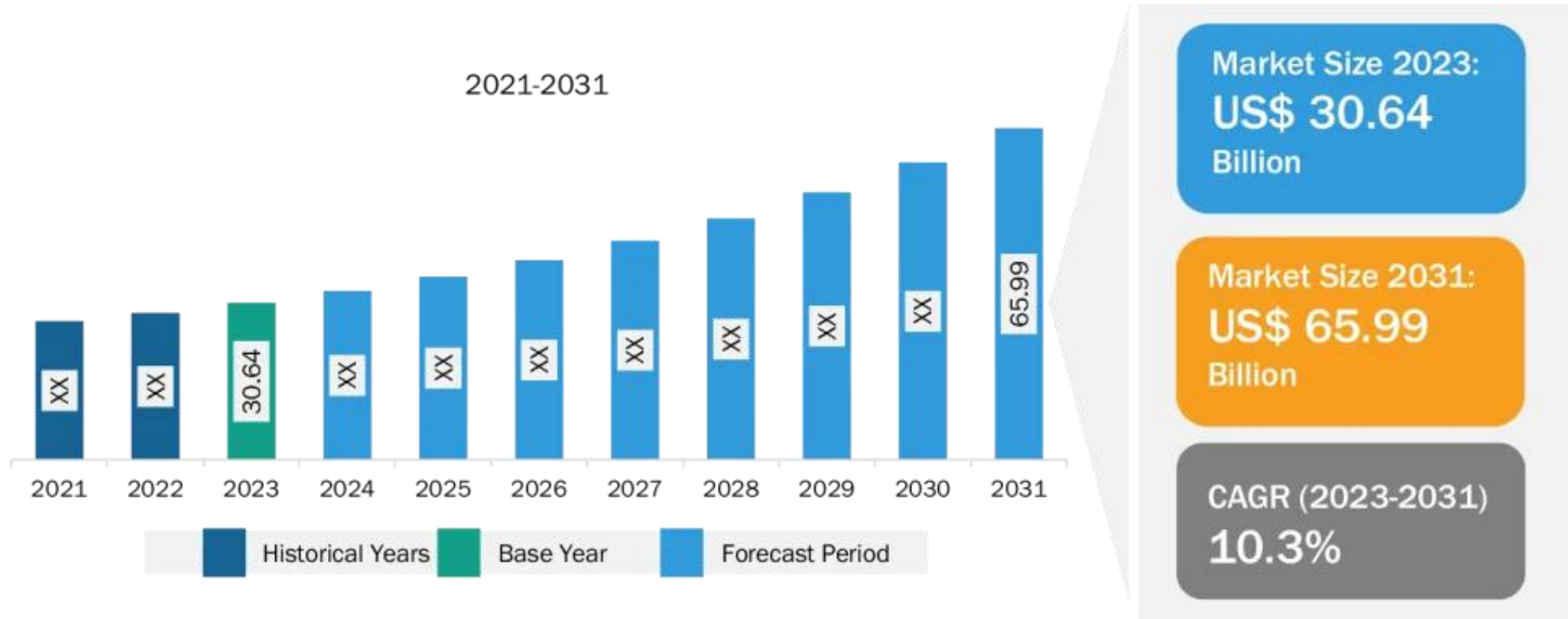


- There are different types of EDI such as Direct EDI, EDI through VAN, EDI through AS2, EDI through FTP/VPN, Web EDI, Mobile EDI, etc.
- VAN and AS2 are the most popular methods.

Does EDI still alive?



EDI forecast



- EDI is an **old** but **still** mainstream technology in **B2B**.

<https://finance.yahoo.com/news/electronic-data-interchange-edi-market-110800334.html>

New format of e-commerce

2019-2020年中国直播电商用户常用直播平台TOP 10



2019-2020年中国直播电商用户常用直播平台TOP 10
TOP 10, Common Live Platform for China's Live E-commerce Users in 2019-2020



数据来源: 艾媒数据中心 (data.iimedia.cn)

样本来源: 草莓派数据调查与计算系统 (Strawberry Pie)

样本量=1983; 调研时间: 2020年1月22日-2月3日

2020H1中国直播电商平台生态链布局及典型案例数据分析报告

艾媒报告中心: report.iimedia.cn ©2020 iiMedia Research In



《真还传》

Making big money

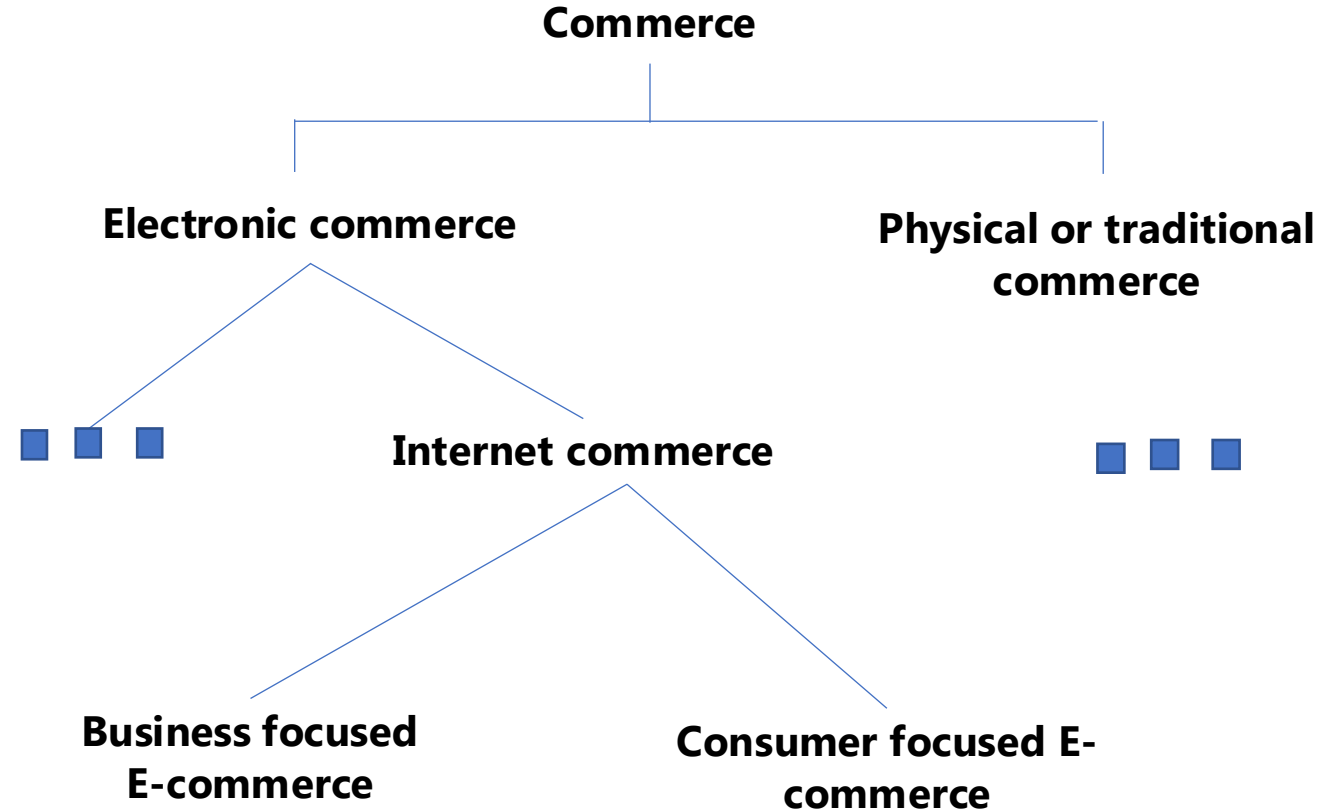
https://www.sohu.com/a/394623312_728793

Chinese Marketplaces serve global markets



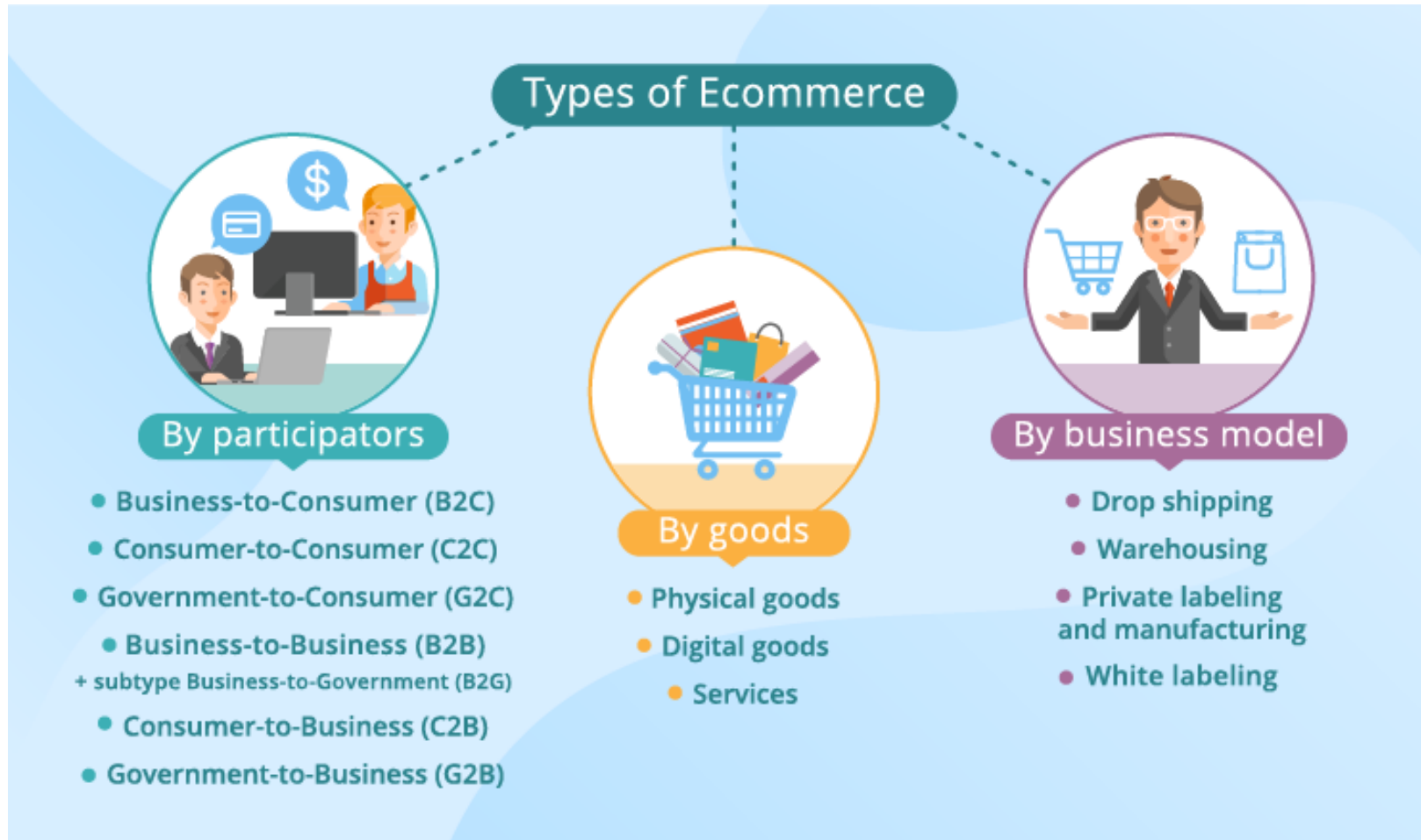
<https://omr.com/en/daily/temu-tiktok-investing-billions>
<https://www.bilibili.com/video/BV1Lj41177Ez/>
<https://www.bilibili.com/video/BV1QM411P7yz/>

Commerce types



- Commerce can be divided into traditional and electronic types.
- There is a number of criteria to classify types of ecommerce.

E-Commerce types



Many terms on different point of view:

O2O (Online + offline) M-B (Mobile business) B2G (Business to government) SoLoMo (Social + local +mobile)

Sharing Economy: P2P B2P

<https://www.scnsoft.com/ecommerce/types-of-ecommerce>

Eight unique features of e-commerce technology

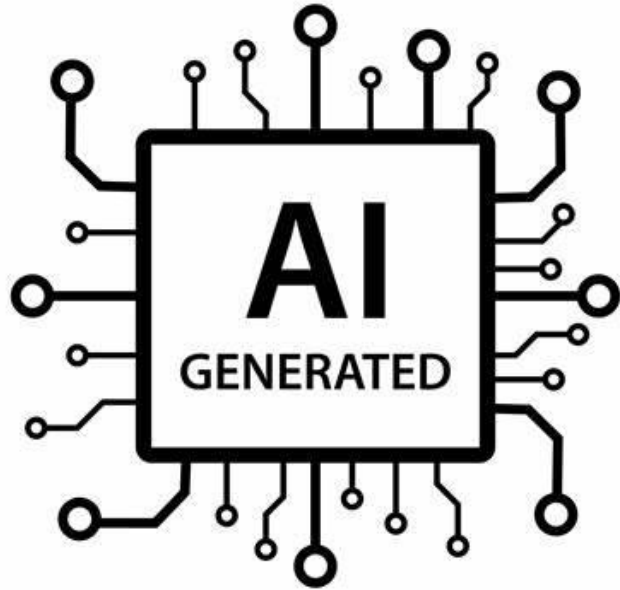
TABLE 2.6 EIGHT UNIQUE FEATURES OF E-COMMERCE TECHNOLOGY	
FEATURE	SELECTED IMPACTS ON BUSINESS ENVIRONMENT
Ubiquity	Alters industry structure by creating new marketing channels and expanding size of overall market. Creates new efficiencies in industry operations and lowers costs of firms' sales operations. Enables new differentiation strategies.
Global reach	Changes industry structure by lowering barriers to entry, but greatly expands market at same time. Lowers cost of industry and firm operations through production and sales efficiencies. Enables competition on a global scale.
Universal standards	Changes industry structure by lowering barriers to entry and intensifying competition within an industry. Lowers costs of industry and firm operations by lowering computing and communications costs. Enables broad scope strategies.
Richness	Alters industry structure by reducing strength of powerful distribution channels. Changes industry and firm operations costs by reducing reliance on sales forces. Enhances post-sales support strategies.
Interactivity	Alters industry structure by reducing threat of substitutes through enhanced customization. Reduces industry and firm costs by reducing reliance on sales forces. Enables differentiation strategies.
Personalization/ Customization	Alters industry structure by reducing threats of substitutes, raising barriers to entry. Reduces value chain costs in industry and firms by lessening reliance on sales forces. Enables personalized marketing strategies.
Information density	Changes industry structure by weakening powerful sales channels, shifting bargaining power to consumers. Reduces industry and firm operations costs by lowering costs of obtaining, processing, and distributing information about suppliers and consumers.
Social technologies	Changes industry structure by shifting programming and editorial decisions to consumers. Creates substitute entertainment products. Energizes a large group of new suppliers.

FIGURE 2.4

HOW E-COMMERCE INFLUENCES INDUSTRY STRUCTURE



Commerce is based on the real products



- Click a mouse **WON'T** generate a car!